- Windows Container Hypothesis Validation - Evidence Triage
- Generated: 2026-11-03 22:24:34 UTC
- Total analyzed issues: 417 (all exported qualifying scored issues; no research cap)
- Observation week: 2026-W45 | first successful observation is frozen.
- Hypothesis 1: Windows container network connectivity barriers; Decision: testing | area: networking | trial week: 3/4 | started: 2026-W43 | trial supporting repositories: 1 | trial contradicting repositories: 0 | fresh this week: 0
- Hypothesis 2: Windows container runtime/host compatibility barriers; Decision: testing | area: runtime/compatibility | trial week: 3/4 | started: 2026-W43 | trial supporting repositories: 1 | trial contradicting repositories: 0 | fresh this week: 0
- Hypothesis 3: The community does not want to use Windows containers as the development platform because it is not free and lacks open source community popularity.; Decision: deferred | area: original | trial week: 0/4 | started: not started | trial supporting repositories: 0 | trial contradicting repositories: 0 | fresh this week: 0
- Hypothesis 4: The Windows containers community has not grown in the past few years because gaps in the Nano Server container image limit what developers can test on top of it.; Decision: deferred | area: original | trial week: 0/4 | started: not started | trial supporting repositories: 0 | trial contradicting repositories: 0 | fresh this week: 0
- Hypothesis 5: No eligible candidate; slot deferred; Decision: deferred | area: unassigned | trial week: 0/4 | started: not started | trial supporting repositories: 0 | trial contradicting repositories: 0 | fresh this week: 0

Hypothesis 1 Supporting evidence

Repository	Source / updated	Title and supporting context	URL	Signal keys
kubernetes/kubernetes	issue | 2026-10-26T13:26:57Z	Dynamic Resource Allocation (DeviceClass/Device) is not supported on Windows nodes Supporting context: ## Why is this needed? DRA exists because the device-plugin + extended-resource model only advertises an integer count of a node-local resource. It cannot express device init/cleanup (e.g. reprogramming an FPGA), partial partitions (Nvidia MIG / SR-IOV), dynamic/network-attached devices, or structured property-based matching. The motivating scenario - device-centric accelerator workloads (GPU/AI, NIC/DPU, edge/batch) - applies to Windows containers exactly as on Linux (Windows GPU/AI is a rea...	https://github.com/kubernetes/kubernetes/issues/141841	networking:connectivity
containerd/nerdctl	issue_comment | 2024-07-24T11:45:54Z	Windows support Supporting context: I must have misremembered about the NAT network limitation then, sorry. I'm sure there was _some_ limitation, but maybe it was in Docker or something? (It may have been in the older virtual network API, too.)	https://github.com/containerd/nerdctl/issues/28#issuecomment-2247702296	networking:connectivity, networking:connectivity@docker

Hypothesis 2 Supporting evidence

Repository	Source / updated	Title and supporting context	URL	Signal keys

Repository	Source / updated	Title and supporting context	URL	Signal keys
kubernetes/website	pull_request | 2026-10-07T16:17:57Z	Document Windows CPU and memory affinity beta graduation Supporting context: - updates the ‘WindowsCPUAndMemoryAffinity‘ feature gate to beta and enabled by default in v1.38; - documents the required CRI, containerd, and runhcs support for Windows CPU and memory affinity; - clarifies Windows Memory Manager ‘BestEffort‘ behavior, CPU and NUMA affinity coordination, and physical memory placement limitations; - documents Memory Manager checkpoint recovery; and - updates the Windows Topology Manager guidance for ‘single-numa-node‘ workloads.	https://github.com/kubernetes/website/pull/57911	runtime:compatibility, runtime:compatibility@containerd
moby/moby	issue | 2026-08-25T00:12:21Z	Flaky test: TestBuildWCOWSandboxSize (insufficient disk space?) Supporting context: ‘‘‘ {"stream":"Step 1/8 : FROM busybox AS intermediate"} {"stream":"\\n"} {"stream":" ---\\u003e ea8f1b6e5c44\\n"} {"stream":"Step 2/8 : WORKDIR C:\\\\\\\\stuff"} {"stream":"\\n"} {"stream":" ---\\u003e Running in a2d453ddd861\\n"} {"stream":"Removing intermediate container a2d453ddd861\\n"} {"stream":" ---\\u003e 395a1b07f269\\n"} {"stream":"Step 3/8 : RUN fsutil file createnew C:\\\\\\\\stuff\\\\\\\\bigfile_0.txt 22548578304 \\u0026\\u0026 del bigfile_0.txt"} {"stream":"\\n"} {"stream":" --...	https://github.com/moby/moby/issues/42743	runtime:compatibility
docker/for-win	issue | 2026-08-10T13:29:36Z	Can’t re-start docker container when AMD GFX in computer (HTTP code 500) server error - hcsshim::ActivateLayer failed in Win32: The process cannot access the fi Supporting context: And I also found that when I install McAfee and when I enable real time virus detection all works fine when I disable real time virus detection "hcsshim::ActivateLayer failed in Win32..." | Can’t re-start docker container when AMD GFX in computer (HTTP code 500) server error - hcsshim::ActivateLayer failed in Win32: The process cannot access the file because it is being used by another process. (0x20) | I want to run windows docker container(--isolation=process) and first time it run just fin...	https://github.com/docker/for-win/issues/14977	runtime:compatibility, runtime:compatibility@docker, runtime:compatibility@hyperv, runtime:compatibility@process

Hypothesis 3 Supporting evidence

Repository	Source / updated	Title and supporting context	URL	Signal keys
No current supporting sources selected				

Hypothesis 4 Supporting evidence

Repository	Source / updated	Title and supporting context	URL	Signal keys

Repository	Source / updated	Title and supporting context	URL	Signal keys
No current supporting sources selected				

Hypothesis 5 Supporting evidence

Repository	Source / updated	Title and supporting context	URL	Signal keys
No current supporting sources selected				

Additional report details

Other report details

- Scored issue export: /home/runner/work/windows-container-issue-tracker/windows-container-issue-tracker/scoring-output/scored-issues.json

Adaptive weekly hypotheses (authoritative latest decisions)

- Observation week: 2026-W45 | first successful observation is frozen.
- Same-week rerun: frozen decision; evidence below is current.
- Hypothesis 1 [networking:connectivity] Windows container network connectivity barriers
- Decision: testing | area: networking | trial week: 3/4 | started: 2026-W43 | trial supporting repositories: 1 | trial contradicting repositories: 0 | fresh this week: 0
- Trial witnesses (historical audit references only, up to 3 per slot; not current showcase):
- kubernetes/kubernetes | issue | 2026-10-21T20:37:12Z
- https://github.com/kubernetes/kubernetes/issues/141841
- Hypothesis 2 [runtime:compatibility] Windows container runtime/host compatibility barriers
- Decision: testing | area: runtime/compatibility | trial week: 3/4 | started: 2026-W43 | trial supporting repositories: 1 | trial contradicting repositories: 0 | fresh this week: 0
- Trial witnesses (historical audit references only, up to 3 per slot; not current showcase):
- kubernetes/kubernetes | pull_request | 2026-10-19T04:26:06Z
- https://github.com/kubernetes/kubernetes/pull/141698
- Hypothesis 3 [original:h1] The community does not want to use Windows containers as the development platform because it is not free and lacks open source community popularity.
- Decision: deferred | area: original | trial week: 0/4 | started: not started | trial supporting repositories: 0 | trial contradicting repositories: 0 | fresh this week: 0
- Trial witnesses (historical audit references only, up to 3 per slot; not current showcase):

- None; no fresh repository support was recorded during this trial.
- Hypothesis 4 [original:h2] The Windows containers community has not grown in the past few years because gaps in the Nano Server container image limit what developers can test on top of it.
- Decision: deferred | area: original | trial week: 0/4 | started: not started | trial supporting repositories: 0 | trial contradicting repositories: 0 | fresh this week: 0
- Trial witnesses (historical audit references only, up to 3 per slot; not current showcase):
- None; no fresh repository support was recorded during this trial.
- Hypothesis 5 [unassigned] No eligible candidate; slot deferred
- Decision: deferred | area: unassigned | trial week: 0/4 | started: not started | trial supporting repositories: 0 | trial contradicting repositories: 0 | fresh this week: 0
- Trial witnesses (historical audit references only, up to 3 per slot; not current showcase):
- None; no fresh repository support was recorded during this trial.

Retired hypotheses

- None yet; no trial has finished.

Researched areas (current evidence, not validated scientific findings)

- Area totals describe research coverage only; trial outcomes require >=3 fresh eligible repositories for the SAME signal, accumulated over its trial.
- networking: 5 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.
- networking:connectivity: eligible supporting repositories=4; fresh=0; contradicting=3; lifecycle: testing in H1.
- networking:connectivity@containerd: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- networking:connectivity@docker: eligible supporting repositories=3; fresh=0; contradicting=2; lifecycle: candidate.
- networking:connectivity@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- networking:connectivity@kubernetes: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- networking:connectivity@ltsc2019: eligible supporting repositories=1; fresh=0; contradicting=1; lifecycle: candidate.
- networking:connectivity@ltsc2022: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- networking:connectivity@ltsc2025: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- networking:connectivity@process: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- networking:dns: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- networking:dns@containerd: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- networking:dns@docker: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- networking:dns@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- networking:dns@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- networking:dns@ltsc2019: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.

- networking:dns@ltsc2025: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity: 2 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.
- identity:authentication: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- identity:authentication@containerd: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- identity:authentication@docker: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- identity:authentication@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:authentication@kubernetes: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- identity:authentication@ltsc2019: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:authentication@ltsc2022: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- identity:authentication@ltsc2025: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- identity:authentication@process: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:gmsa: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:gmsa@containerd: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:gmsa@docker: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:gmsa@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- identity:gmsa@ltsc2022: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images/Nano: 2 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.
- images:nano-api: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- images:nano-api@containerd: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:nano-api@docker: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- images:nano-api@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:nano-api@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:nano-api@ltsc2019: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:nano-api@ltsc2022: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- images:nano-api@ltsc2025: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- images:nano-api@process: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- images:size: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@containerd: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@docker: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@ltsc2019: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.

- images:size@ltsc2022: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@ltsc2025: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- images:size@process: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.

runtime/compatibility: 4 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.

- runtime:compatibility: eligible supporting repositories=4; fresh=0; contradicting=4; lifecycle: testing in H2.
- runtime:compatibility@containerd: eligible supporting repositories=3; fresh=0; contradicting=0; lifecycle: candidate.
- runtime:compatibility@docker: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- runtime:compatibility@hyperv: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- runtime:compatibility@kubernetes: eligible supporting repositories=1; fresh=0; contradicting=1; lifecycle: candidate.
- runtime:compatibility@ltsc2019: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- runtime:compatibility@ltsc2022: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- runtime:compatibility@ltsc2025: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- runtime:compatibility@process: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.

tooling/build: 5 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.

- tooling:build: eligible supporting repositories=5; fresh=0; contradicting=1; lifecycle: candidate.
- tooling:build@containerd: eligible supporting repositories=2; fresh=0; contradicting=0; lifecycle: candidate.
- tooling:build@docker: eligible supporting repositories=2; fresh=0; contradicting=1; lifecycle: candidate.
- tooling:build@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- tooling:build@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- tooling:build@ltsc2019: eligible supporting repositories=3; fresh=0; contradicting=0; lifecycle: candidate.
- tooling:build@ltsc2022: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.
- tooling:build@ltsc2025: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- tooling:build@process: eligible supporting repositories=1; fresh=0; contradicting=0; lifecycle: candidate.

licensing/cost: 0 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.

- licensing:cost: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- licensing:cost@docker: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- licensing:cost@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- licensing:cost@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- licensing:cost@ltsc2022: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- licensing:cost@ltsc2025: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- licensing:cost@process: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.

community: 0 distinct eligible supporting repositories; 0 fresh. Area totals are not qualifications or validations.

- community:adoption: eligible supporting repositories=0; fresh=0; contradicting=1; lifecycle: no eligible support.
- community:adoption@docker: eligible supporting repositories=0; fresh=0; contradicting=1; lifecycle: no eligible support.
- community:adoption@hyperv: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- community:adoption@kubernetes: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- community:adoption@ltsc2019: eligible supporting repositories=0; fresh=0; contradicting=1; lifecycle: no eligible support.
- community:adoption@ltsc2022: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- community:adoption@ltsc2025: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.
- community:adoption@process: eligible supporting repositories=0; fresh=0; contradicting=0; lifecycle: no eligible support.

Adaptive method and limitations

- Each hypothesis is tested for 4 weekly observations; only fresh eligible evidence
- observed during the trial accumulates, by distinct repository, across those weeks.
- Trial outcome: supported = >=3 supporting repositories and more supporting than
- contradicting; contradicted = contradicting repositories present and >= supporting; otherwise inconclusive.
- Finished trials retire, free their slot the same week and cannot be retested for 8 weeks.
- Candidates rank by fresh supporting repositories, then untested before previously tested, then
- all-time eligible supporting repositories, unique issue URLs (not comments/PR boosts), seed
- hypotheses before other signals, and signal id.
- Historical sources are discovery only: they nominate candidates but never count as trial evidence.
- Overlapping hypotheses (same base signal, or a seed and its aliases) are never tested together.
- Context signals (signal@context) refine a base signal with Windows version, isolation or
- orchestrator terms found in the same paragraph or source title; they are narrower hypotheses.
- This is deterministic evidence triage, not scientific proof, validation or causal inference.
- Independent means distinct full repository names, NOT distinct owners or independent people.
- Only API isPrivate=false sources count; Microsoft/AKS sources remain context only.
- Excluded owners: aks, azure, azure-devops, azure-samples, azuread, dotnet, microsoft, microsoft-docker, microsoftdocs, microsoftgraph, powershell, windows, windowsserverdocs; additional owners: HYPOTHESIS_EXCLUDED_OWNERS.
- Current evidence window (UTC): (2026-10-27T22:24:34.265061+00:00, 2026-11-03T22:24:34.265061+00:00].
- Freshness uses source/comment updatedAt in the trailing 7 days, excluding the lower boundary.
- For consecutive weekly observations, the window starts no earlier than the prior observed_at.
- This prevents overlapping evidence reuse; missed weeks contribute no evidence for the gap.
- ISO weeks freeze decisions, not source dates. Future dates never count.
- Issue/PR updatedAt may reflect unrelated metadata, not a changed claim; comment dates are source-specific.

- Only latest 100 comments and most recently updated linked PR are examined.
- Taxonomy: area + explicit barrier intent; specific DNS/gMSA/Nano intents take precedence.
- Negation/resolution/non-barriers are separate contrary/inconclusive evidence, never supporting votes.
- Signal taxonomy: networking:dns = DNS resolution failures; networking:connectivity = network connectivity barriers; identity:gmsa = gMSA authentication barriers; identity:authentication = authentication barriers; images:nano-api = Nano Server API/testing limitations; runtime:compatibility = runtime/host compatibility barriers; images:size = image size/distribution barriers; tooling:build = build/tooling failures; licensing:cost = licensing/cost barriers; community:adoption = community/adoption barriers
- Context qualifiers: @ltsc2019 = on Windows Server 2019/LTSC 2019; @ltsc2022 = on Windows Server 2022/LTSC 2022; @ltsc2025 = on Windows Server 2025/LTSC 2025; @hyperv = involving Hyper-V; @process = with process isolation; @kubernetes = on Kubernetes; @containerd = on containerd; @docker = on Docker

Current supporting source showcase (separate from frozen historical decision references)

- Supporting sources shown: 5 of 49 eligible unique URLs (maximum 5 globally).
- Selection uses exclusively verified public, non-Microsoft/non-excluded-owner sources.
- Source rank: current hypothesis key first; urgency label; recent discussion; updatedAt; URL.
- Urgency (case-insensitive exact labels): p0/critical/blocker > p1/high/urgent >
- p2/medium > p3/low > p4/lowest > unknown; optional 'priority:'/'priority/'
- or 'severity:'/'severity/' prefixes. No urgency inferred from prose or issue score.
- Diversity first: best ranked source per distinct repository, then ranked fills to five.
- Discussion = unique comments created in (now - 7 days, now], among latest 100 fetched;
- edits and future-created comments do not add activity. No author diversity is inferred.
- Comment urgency/activity use their parent issue/PR; source repository is always its own.
- Other evidence - contradicting: 59 candidate signals across 26 unique source URLs (all researched sources, including excluded context; not supporting votes).
- Other evidence - inconclusive: 2841 candidate signals across 368 unique source URLs (all researched sources, including excluded context; not supporting votes).

Original seed hypotheses (historical reference only)

- Seed 1: The community does not want to use Windows containers as the development platform because it is not free and lacks open source community popularity.
- Seed 2: The Windows containers community has not grown in the past few years because gaps in the Nano Server container image limit what developers can test on top of it.

Method and limitations

- Fresh issue title/body/state/updatedAt/URL, first 100 labels and latest 100 comments; all pages of
- cross-referenced and connected timeline events are inspected for actual PR nodes.

- Only the most recently updated linked PR is examined (ties: node id), including
- its current title/body/state/mergedAt and latest 100 comments. References are not endorsements.
- Requests are serial via the shared rate-aware GitHub client. No external LLM or
- arbitrary URLs from issue text are fetched. API/missing-data errors abort the report.
- Deterministic paragraph-level keyword rules produce supporting, contradicting and
- inconclusive candidate signals, not automated semantic truth or hypothesis proof.
- Original seed 1 lexical support requires Windows containers, cost, community, development and
- barrier terms together without detected negation, except affirmative ’not free’
- cost wording. Other negation, including ’not a barrier’ or ’do not avoid’, remains
- disqualifying for seed 1 support. Original seed 2 lexical support requires
- Nano Server gap/testing terms without detected negation. Explicit non-barrier/use
- or resolved-gap phrases can be contradicting candidates; other matches are inconclusive.
- Rules can miss paraphrases, sarcasm, code, quotations, historical context and negation;
- unrelated Windows needs and topic mentions are not definitive support.
- Triage retains paragraph context up to 1800 characters; showcase aggregates supporting
- signal keys/excerpts by source URL into at most 500 characters, with truncation marked.
- Source dates are last-updated timestamps, not necessarily when a claim was made.
- Comments older than the latest 100 and non-selected PR contents are outside scope.
- A scored issue sample cannot establish prevalence, causality, community growth,
- or represent nonadopters. Activity, scores, open/closed/merged states do not
- establish popularity or growth. This report does not determine licensing/free-use truth.

Research limitations

- Do not interpret candidate counts as prevalence or validation. Review full sources
- and obtain representative adopter/nonadopter research and longitudinal growth data.